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Allergic rhinitis: Clinical manifestations, epidemiology, pathogenesis, and diagnosis

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INTRODUCTION AND TERMINOLOGY

Allergic rhinitis, or allergic rhinosinusitis, is characterized by inflammation of the nasal mucosa leading to paroxysms of sneezing, rhinorrhea, and nasal obstruction, often accompanied by itching of the eyes, nose, and palate [1]. Postnasal drip, cough, irritability, and fatigue are other common symptoms [2-4].

Some investigators prefer the term "rhinosinusitis" to the separate terms "rhinitis" and "sinusitis." This is because the nose and sinus mucosa are contiguous, rhinitis and sinusitis frequently occur together, rhinitis commonly leads to sinusitis, and nasal symptoms are common with sinusitis. However, within this topic review, rhinitis and sinusitis are referred to separately given that management issues may differ for each condition, and detailed reviews of acute and chronic sinusitis are presented elsewhere. (See "Acute sinusitis and rhinosinusitis in adults: Clinical manifestations and diagnosis" and "Chronic rhinosinusitis: Clinical manifestations, pathophysiology, and diagnosis" and "Uncomplicated acute sinusitis and rhinosinusitis in adults: Treatment".)

The clinical manifestations, epidemiology, pathogenesis, and diagnosis of allergic rhinitis are presented in this topic review. The treatment of this condition is discussed separately. (See "Pharmacotherapy of allergic rhinitis".)

EPIDEMIOLOGY

Allergic rhinitis is a common condition affecting 10 to 30 percent of children and adults in the United States and other resource-abundant countries [5-10]. It may be less common in some

parts of the world, although even resource-limited countries report significant rates [11-17]. There is a higher incidence of allergic rhinitis and allergic diseases in general in urban and industrialized areas when compared with rural areas [18-21].

The prevalence of rhinoconjunctivitis, asthma, and eczema were systematically evaluated in approximately 1.2 million children in 98 countries in the International Study of Asthma and Allergies in Childhood (ISAAC) [22]. The overall prevalence of rhinoconjunctivitis in children aged 6 to 7 years and 13 to 14 years was 8.5 and 14.6 percent, respectively. A subsequent meta-analysis demonstrated a prevalence of self-reported lifetime allergic rhinitis of 19.93 percent in children [23]. Allergic rhinitis is more common in males in childhood, but this changes to a female predominance in adolescence [24].

Multiple studies have shown an increase in the prevalence of allergic rhinitis over time [23]. As an example, in a Chinese study analyzing three random-digit dialing phone surveys performed in 2005, 2011, and 2019, and including over 25,000 people in 27 cities, the average prevalence of allergic rhinitis per city increased from 11 to 18 to 21 percent over those 14 years [25]. Worldwide, increasing prevalence is thought to be partly due to climate change, leading to longer pollen seasons and higher pollen counts [26-28]. The impact of climate change on allergens (including pollen) is discussed in detail elsewhere. (See "Overview of health effects of climate change", section on 'Allergens'.)

Economic burden — Allergic rhinitis is associated with significant morbidity and expense [29-34]:

- It accounts for at least 2.5 percent of all clinician visits, 2 million lost school days, 6 million lost work days, and 28 million restricted work days per year.
- The average number of annual prescriptions for patients with allergic rhinitis is nearly double that for patients without allergic rhinitis (19 versus 10) [33].
- Prescription medications account for almost half of the direct medical costs of rhinitis [35,36].

Economic burden secondary to allergic rhinitis is significant. Costs related to this condition are approximately $3.4 billion in the United States, most of which are associated with medication costs [37]. Medical spending to treat allergic rhinitis almost doubled from 2000 to 2005 (USD $6.1 to $11.2 billion) [38]. In addition to costs directly attributable to allergic rhinitis, the disorder is highly associated with asthma and sinusitis, further expanding its economic impact.

PATHOGENESIS

Allergens contain proteases that activate and damage nasal epithelial cells, which then secrete alarmins (eg, interleukin [IL] 25, IL-33, thymic stromal lymphopoietin [TSLP], and other damage-associated molecular patterns [DAMPs]). In predisposed individuals, these signals initiate a T helper type 2 (Th2) type inflammatory response that results in production of allergen-specific immunoglobulin E (sIgE) or "sensitization" to that allergen [38,39]. (See "The biology of IgE".)

After IgE antibodies specific for a certain allergen are synthesized and secreted, they bind to high-affinity receptors on mast cells (and basophils). When an allergen is inhaled into the nose, it cross-links these allergen-specific cell-bound IgE antibodies on the mast cell surface, resulting in rapid degranulation and mediator release. These mediators are either preformed, associated with granules, formed during degranulation, or generated after transcription [40]. (See "Mast cell-derived mediators".)

Early- and late-phase responses — The allergic response consists of an early and late phase (figure 1).

- The early phase begins within seconds and peaks at approximately 30 minutes. Histamine is the most important preformed mediator in allergic rhinitis. Histamine reproduces **all** of the early-phase symptoms of allergic rhinitis when sprayed into the noses of normal volunteers. Histamine concentrations in nasal secretions correlate temporally with symptoms (sneezing, nasal pruritus) and increases in nasal airway resistance (congestion) (figure 2) [41,42]. Mast cell activation also stimulates the production of other mediators during degranulation, including platelet-activating factor (PAF), bradykinin, prostaglandins, and leukotrienes. The superficial nasal epithelium in patients with allergic rhinitis has 50-fold more mast cells and basophils per specimen than does epithelium from nonallergic patients.

- A late-phase nasal allergic reaction develops in approximately 50 percent of patients with seasonal rhinitis, which peaks at 6 to 12 hours after nasal allergen challenge [43]. This corresponds to an influx of basophils, neutrophils, mononuclear cells, and eosinophils [44,45]. Tissue eosinophilia is also characteristic of allergic rhinitis [43]. This secondary inflammatory response is thought to be important in establishing the chronicity of the disorder.

Involvement of the paranasal sinuses — Allergic rhinitis is more accurately described as allergic rhinosinusitis because the inflammatory response noted in the mucosa of patients with allergic rhinitis often extends to the paranasal sinuses as well [46].

RISK FACTORS

The following are proposed or identified risk factors for allergic rhinitis [17,47-53]:

- Personal or family history of atopy (ie, the genetic predisposition to develop allergic diseases) [50,54,55]
- Male sex
- Birth during the pollen season
- Firstborn status
- Early use of antibiotics [56]
- Maternal smoking exposure in the first year of life
- Exposure to indoor allergens, such as dust mite and molds
- Presence of allergen-specific IgE (sIgE) [53]
- Serum IgE >100 international units/mL before age six years
- Frequent respiratory tract infections or long-lasting infections
- Tonsillectomy
- Heavy traffic near the home
- Living near a weedy area, an air-polluting factory, or a mine

A review of multiple studies reported that the presence of each of these factors was associated with a positive likelihood ratio for the diagnosis that ranged from three to five [57].

CLINICAL MANIFESTATIONS

Signs and symptoms — Allergic rhinitis presents with paroxysms of sneezing, rhinorrhea, nasal obstruction, and nasal itching. Postnasal drip, cough, irritability, and fatigue are other common symptoms [2-4]. Some patients experience itching of the palate and inner ear. Associated ocular symptoms include bilateral itching, tearing, and/or burning of the eyes in patients with concomitant allergic conjunctivitis. (See "Allergic conjunctivitis: Clinical manifestations and diagnosis", section on 'Clinical manifestations'.)

Young children typically do not blow their noses, and instead, they may repeatedly snort, sniff, cough, and clear their throats. Severe nasal congestion can lead to snoring and obstructive sleep apnea. Some scratch their itchy palates with their tongues, producing a clicking sound (palatal click).

Quality of life and cognitive function — Sleep-disturbed breathing is one of the most important sequelae of untreated allergic rhinitis [58,59]. Sleep quality in general is disturbed by allergic rhinitis [52]. Fatigue and generalized malaise are common, although patients rarely report these symptoms directly [60,61]. Allergic rhinitis is associated with a host of cognitive and psychiatric issues in children and adolescents, including attention deficit hyperactivity disorder, lower exam scores during peak pollen seasons [62], poor concentration, impaired athletic performance, and low self-esteem [9,63-65]. In adults, allergic rhinitis is associated with anxiety, depression, reduced academic performance and

work productivity (and lower than that of patients with asthma), impaired sexual performance, and lower quality-of-life scores [11,60,66-69].

Patterns of symptoms — Allergic rhinitis may be classified by temporal pattern (intermittent or persistent) and by severity (mild or moderate-severe) (table 1) [70,71]:

- **Intermittent** – Symptoms are present less than four days per week or for less than four weeks.
- **Persistent** – Symptoms are present more than four days per week and for more than four weeks.
- **Mild** – None of the items listed below for "moderate-severe" are present.
- **Moderate-severe** – One or more of the following items is present:
  - Sleep disturbance
  - Impairment of school or work performance
  - Impairment of daily activities, leisure, and/or sport activities
  - Troublesome symptoms

This classification system was proposed by an international workshop of 34 specialists in respiratory allergy in collaboration with the World Health Organization (WHO) [70]. The WHO had targeted allergic rhinitis because of its impact on asthma and mirrors the consensus asthma guidelines.

Other commonly used terms are "seasonal," which is allergic rhinitis that occurs at a particular time of the year (figure 3), and "perennial," which describes symptoms to allergens that are present year-round [72]. This system of classification is preferred by the US Food and Drug Administration (FDA).

Patients whose symptoms occur episodically are often more aware of the disability caused by allergic rhinitis, whereas patients with chronic symptoms often adapt to significant impairment over time and may not seek medical care until symptoms become severe [64,65,73,74]. Children, in particular, will endure significant disability.

Persistent/perennial symptoms are more common than purely intermittent or seasonal symptoms, although many patients have perennial symptoms with seasonal exacerbations [2,75].

Seasonal allergic rhinitis is usually caused by pollen from trees, grasses, and weeds. Depending upon the geographic area, pollination periods for certain types of plants are well known (figure 3). Colloquial names sometimes correctly identify the triggering pollen (eg, cedar fever), although, in other cases, a plant that is not causing symptoms but

is very visible at the same time is implicated by the name (eg, rose fever, which refers to allergic rhinitis caused by grasses that pollinate at the same time that roses bloom or hayfever, which occurs in the fall when hay is being gathered but is caused by weed pollens or from mold growing in the hay). Symptoms of seasonal allergic rhinitis are predictable and reproducible from year to year.

Perennial allergic rhinitis usually reflects allergy to indoor allergens such as dust mites, cockroaches, mold spores, or animal dander, although aeroallergens may cause perennial rhinitis in tropical or subtropical climates. Perennial rhinitis due to outdoor allergens is common in subtropical regions with long pollinating seasons and ubiquitous mold and dust mite allergens. Perennial symptoms are also seen with occupational allergen exposure [76]. (See "Occupational rhinitis".)

It is important to understand that allergic rhinitis, whether seasonal or perennial, may be difficult to distinguish clinically from the nonallergic forms of rhinitis since not all seasonal and perennial symptoms are unique to allergic forms of rhinitis [2,75]. As an example, chronic nonallergic rhinitis can be triggered by changes in weather and temperature and may appear to have a seasonal pattern in some patients. Thus, for an accurate diagnosis of allergic rhinitis, diagnostic testing may be required. (See 'Allergen-specific testing' below and "Chronic nonallergic rhinitis".)

Increasing sensitivity over time — When a patient is continually exposed to an allergen, persistent nasal mucosal inflammation develops. In such patients, symptoms of rhinitis occur on exposure to lower doses of allergen (priming) and to nonspecific irritants (hyperreactivity). Clinically, this results in continued and frequently more severe rhinitis symptoms with exposure to low allergen concentrations. This phenomenon of increasing sensitivity over time probably results from a lowering of the threshold for a clinical response. Sequential allergen challenges in patients with allergic rhinitis result in more symptoms and higher levels of histamine and inflammatory cells in nasal washes [77]. Presumably, the induced inflammation recruits additional inflammatory cells and their products into the process, producing increasing symptoms.

Concomitant with allergen priming, the nose becomes progressively more sensitive to cholinomimetic stimuli, such as methacholine and histamine, as well as irritant stimuli like cold air. Over time, many patients with allergic rhinitis report heightened sensitivity to irritants (eg, tobacco smoke, particulate pollution), volatile substances, and strong scents and perfumes [78].

Physical findings — The following physical findings may be present in patients with active allergic rhinitis [79-82]:

- Infraorbital edema and darkening due to subcutaneous venodilation, findings that are sometimes referred to as "allergic shiners."

- Accentuated lines or folds below the lower lids (Dennie-Morgan lines), which suggest concomitant allergic conjunctivitis. (See "Allergic conjunctivitis: Clinical manifestations and diagnosis".)

- A transverse nasal crease caused by repeated rubbing and pushing the tip of the nose up with the hand (the "allergic salute").

- "Allergic facies," which are typically seen in children with early-onset allergic rhinitis, consist of a highly arched palate, open mouth due to mouth breathing, and dental malocclusion (figure 4).

The internal structures of the nose, oropharynx, and ears should be examined:

- The nasal mucosa of patients with active allergic rhinitis frequently has a pale bluish hue or pallor along with turbinate edema.

- Clear rhinorrhea may be visible anteriorly, or, if the nasal passages are obstructed, rhinorrhea may be visible dripping down the posterior pharynx.

- Hyperplastic lymphoid tissue lining the posterior pharynx, which resembles cobblestones (a finding called "cobblestoning").

- Tympanic membranes may retract, or serous fluid may accumulate behind tympanic membranes in patients with significant nasal mucosal swelling and eustachian tube dysfunction [83].

Routine laboratory findings — Routine laboratories are usually normal. Neither peripheral blood eosinophil counts nor total serum IgE levels (elevated in only 30 to 40 percent of patients) are sensitive enough to help diagnose allergic rhinitis. (See "The relationship between IgE and allergic disease", section on 'Allergic rhinitis'.)

Natural history — Allergic rhinitis typically requires a few years of allergen exposure to develop. In a study of approximately 600 children, at least two seasons of pollen exposure were required before most children developed pollen-related allergic symptoms [84]. Accordingly, it is uncommon in children under two years of age. If a very young child appears to have persistent nasal symptoms, other disorders should be considered. (See 'Differential diagnosis' below.)

Sensitization to aeroallergens generally precedes the appearance of rhinitis symptoms [17,85]. Sensitization describes the presence of allergen-specific IgE as measured by skin testing or in vitro tests. However, sensitization is not synonymous with allergy, and a person can be sensitized to an allergen without developing symptoms when exposed to that allergen. Only a subset of sensitized individuals demonstrate clinical allergy [86]. The distinction

between sensitization and clinical allergy is discussed in more detail elsewhere. (See "The relationship between IgE and allergic disease", section on 'Sensitization'.)

In children, sensitization and then clinical allergy develop first to allergens that are continually present in the environment (eg, dust mites or animal danders) and then to pollens and other seasonal allergens.

After the age of two years, the prevalence of allergic rhinitis steadily increases, demonstrating a bimodal peak in the early school and early adult years. The condition is usually persistent throughout adulthood, with some improvement in older age [30,87-92].

- In a prospective study of 2024 children, the prevalence of allergic rhinitis increased from 5 percent at the age of four years to 14 percent at age eight years [17,85]. The prevalence then gradually increases, peaking in early adulthood.

- In a 20-year prospective study of 1137 patients followed from age 4 to 24 years, the prevalence of pollen-induced allergic rhinitis increased from 13 to 35 percent, although most of the increase occurred by age 16 years, with minimal increase from 16 to 24 years [53]. Approximately three-quarters of children had persistent allergic rhinitis, and one-third developed asthma.

Allergic rhinitis uncommonly presents for the first time in older adults, unless there is a significant change in exposures (eg, a new pet or a move to a different climate). Thus, in an older adult with new nasal symptoms, other causes of rhinitis should be considered. A change in response to treatment in older adults may indicate the development of vasomotor or atrophic rhinitis. (See "Atrophic rhinosinusitis".)

Data on remission are limited, although it appears to be uncommon in childhood and adolescence. However, rates of remission as high as 30 percent have been reported in adults [93-95].

ASSOCIATED CONDITIONS

Allergic rhinitis occurs in association with a number of other disorders, including allergic conjunctivitis, acute or chronic sinusitis, asthma, and atopic dermatitis (eczema).

- **Allergic conjunctivitis** – Up to 60 percent of patients with allergic rhinitis have concomitant allergic conjunctivitis [96,97]. Allergic conjunctivitis presents with itching, tearing, conjunctival edema, hyperemia, watery discharge, burning, and photophobia (picture 1). Eyelid edema is also common. Symptoms are usually bilateral. (See "Allergic conjunctivitis: Clinical manifestations and diagnosis".)

- **Sinusitis** – Nasal inflammation associated with allergic rhinitis can also cause obstruction of the sinus ostiomeatal complex, thereby predisposing to bacterial infection of the sinuses. This process accounts for as much as 30 to 80 percent of cases of acute and chronic bacterial sinusitis, respectively. However, purulent sinusitis without concurrent rhinitis is rare given that these tissues are anatomically contiguous [98].

  Symptoms of bacterial sinusitis include nasal congestion, purulent rhinorrhea or postnasal drip, facial or dental pain, and cough. Purulent rhinorrhea, purulent postnasal drip, pain in a maxillary tooth, or, in children, persistent cough are the most useful predictors of bacterial sinusitis. However, no single symptom has a high degree of sensitivity or specificity in discriminating bacterial sinusitis from allergic or viral rhinitis. (See "Acute sinusitis and rhinosinusitis in adults: Clinical manifestations and diagnosis" and "Acute bacterial rhinosinusitis in children: Clinical features and diagnosis", section on 'Anatomy'.)

- **Asthma** – Up to 40 percent of adults with asthma have allergic rhinitis [99]. In children, cough and wheeze are the most common symptoms and are often more prominent with exertion (table 2). Adults may report any combination of cough, wheeze, or chest tightness. Airway hyperresponsiveness (AHR) is common in individuals with allergic rhinitis, even in the absence of a history of wheezing [100]. Allergic rhinitis is a risk factor for the development of asthma in both children and adults [101-103]. (See "Epidemiology of asthma" and "Asthma in children younger than 12 years: Initial evaluation and diagnosis" and "Asthma in adolescents and adults: Evaluation and diagnosis".)

- **Atopic dermatitis (eczema)** – In children, atopic dermatitis presents with intensely pruritic erythematous patches with papules and some crusting, usually affecting the face, scalp, extremities, or trunk, with sparing of the diaper areas.

  In older individuals with more chronic disease, atopic dermatitis presents with thickened skin, increased skin markings (lichenification), and excoriated and fibrotic papules. The flexural areas (neck, antecubital fossae, and popliteal fossae) are most commonly involved in adults. This disorder is discussed separately. (See "Atopic dermatitis (eczema): Pathogenesis, clinical manifestations, and diagnosis".)

- **Pollen-food allergy syndrome (also known as oral allergy syndrome)** – Pollen-food allergy syndrome (PFAS), sometimes referred to as "oral allergy syndrome" is a form of food allergy that develops in individuals who are sensitized to pollens. Patients report itching and/or mild swelling of the mouth and throat immediately following ingestion of certain uncooked fruits (such as apples, peaches, plums, cherries, and some nuts) or raw vegetables. The symptoms result from cross-reactivity between specific IgE to pollen-related proteins in these foods. (See "Pollen-food allergy syndrome (oral allergy syndrome): Clinical manifestations and diagnosis" and "Pollen-food allergy syndrome (oral allergy syndrome): Management and prognosis".)

- **Other conditions** – Allergic rhinitis is strongly associated and probably causally related to eustachian tube dysfunction, causing concomitant serous and acute otitis media [83]. Nasal obstruction due to severe allergic rhinitis can also cause sleep-disordered breathing and anosmia [104,105]. There may be an increased prevalence of migraine headache in patients with allergic rhinitis [106].

DIAGNOSIS

The diagnosis of allergic rhinitis can be made on **clinical grounds** based upon the presence of characteristic symptoms (ie, paroxysms of sneezing, rhinorrhea, nasal obstruction, nasal itching, postnasal drip, cough, irritability, and fatigue), a suggestive clinical history (including the presence of risk factors), and supportive findings on physical examination [107,108]. Skin prick testing confirms that the patient is sensitized to aeroallergens, although it is not necessary for the initial diagnosis. Serum IgE testing can be used as a reasonable alternative for skin prick testing in situations where patients cannot undergo the latter (eg, severe dermatologic condition) [52,108,109].

There is insufficient evidence to routinely support other testing [108]. Imaging is not usually performed in the diagnosis of allergic rhinitis unless a concomitant condition such as chronic rhinosinusitis (CRS) is suspected or there is a history of facial trauma or features to suggest anatomic abnormalities (unilateral congestion or obstruction).

If the patient's symptoms prove difficult to manage or the trigger(s) for the symptoms are not apparent, then further evaluation is indicated to demonstrate that the patient is sensitized to aeroallergens and that symptoms occur when expected for the allergens in question. Sensitization can be demonstrated with either allergy skin testing or in vitro tests for allergen-specific IgE.

A positive response to a therapeutic trial of either topical nasal corticosteroids or topical antihistamines does not conclusively establish a diagnosis of allergic rhinitis, because these therapies are also effective in the treatment of nonallergic rhinitis [2,110]. (See "Chronic nonallergic rhinitis", section on 'Management'.)

History — Some forms of allergic rhinitis can be readily diagnosed by history alone, while others may require additional evaluation for accurate diagnosis:

- Seasonal allergic rhinitis is often diagnosed by the history alone because it is reproducible from year to year. Seasonal allergic rhinitis caused by tree and grass pollen typically occurs in the spring and summer, and symptoms caused by ragweed pollen exposure occur in the fall, although there are regional variations (figure 3).

- Some cases of episodic allergic rhinitis can be diagnosed by history alone if there is an obvious connection between exposure and the onset of symptoms. As an example, a history of episodic exposure to animals or high levels of house dust resulting in acute allergic symptoms may be readily diagnosed as episodic allergic rhinitis.

- By comparison, the culprit allergens in perennial/persistent allergic rhinitis may not be readily apparent from the clinical history. Perennial allergic rhinitis usually reflects allergy to indoor allergens like dust mites, cockroaches, or animal dander, although pollens may cause perennial rhinitis in tropical or subtropical climates.

A British study of 143 adult subjects evaluated the ability of specific questions to identify patients in whom subsequent skin testing for common aeroallergens was likely to be negative. Among patients who answered "no" to all four of the below questions, 88 percent had negative results on skin prick testing to house dust mite, mixed grass pollen, and cat and dog aeroallergens [111].

- Do you have or have you ever had hayfever?
- Do your parents or any of your siblings have or have they ever had hayfever?
- Do your allergy symptoms vary when you go from place to place (eg, on holiday)?
- Is there a specific trigger that always sets off your allergy symptoms?

For populations living in similar temperate regions of the world, this may be a useful tool for identifying patients who are less likely to benefit from an allergic diagnostic evaluation, although it may not apply to all ages and populations (eg, geographic, climatic, and phenotypic/genotypic heterogeneity) and should be validated in larger numbers of people.

Physical examination — The nose, oropharynx, tympanic membranes, and eyes should be examined as each of these structures may show findings of allergic rhinitis or associated disorders. (See 'Physical findings' above.)

The internal structures of the nose and the nasal mucosa can be visualized using a standard office otoscope with a disposable tip. Stabilizing the tip against the patient's upper nares prevents painful prodding of the mucosa, which is normally exquisitely sensitive to touch.

In patients older than five years of age, flexible fiberoptic rhinoscopy is helpful but not essential for diagnosis. If pursued, it should be performed by clinicians specifically trained in the technique. This technique is described elsewhere. (See "Chronic rhinosinusitis: Clinical manifestations, pathophysiology, and diagnosis", section on 'Rhinoscopy and endoscopy'.)

Allergen-specific testing — It is not necessary to perform testing for allergen-specific IgE either with blood tests or skin testing before making the presumptive diagnosis of allergic rhinitis and initiating treatment. Primary care clinicians treat the majority of patients with

allergic rhinitis and often initiate therapy empirically, identifying possible triggers only through the clinical history. This approach is adequate for many patients.

Indications — Although identification of the allergens to which the patient is sensitized is not needed in all cases, the use of diagnostic testing has been associated with improved symptom control and quality of life [112]. Identifying the allergens that are important to an individual facilitates avoidance of the allergen and identifies candidates for allergen immunotherapy, which can eventually reduce reliance on chronic medications. Patients who wish to avoid medications and want to maximize allergen avoidance are candidates for evaluation by an allergist. Other indications for early referral to an allergist are patients with symptoms that interfere with performance at school or work.

Skin testing — Immediate hypersensitivity skin testing (prick skin tests) is a quick and cost-effective way to identify the presence of allergen-specific IgE [113-115]. These tests are performed by allergy specialists because, although generally considered quite safe, rare systemic allergic reactions are possible in response to the testing itself. In addition, an allergist can properly pair the skin prick test results to the clinical history and better interpret indeterminate pick test responses (eg, in the setting of concomitant dermographism). In sensitive patients, testing with selected diagnostic solutions of tree, grass, or weed pollen, mold, house dust mite, and/or animal allergens results in a wheal-and-flare reaction at the skin test site within 15 to 20 minutes [115]. Positive prick skin tests correlate more closely with symptoms than intradermal tests. (See "Overview of skin testing for IgE-mediated allergic disease".)

Skin testing is particularly useful among patients with:

- An unclear diagnosis based upon the history and physical examination
- Poorly controlled symptoms, such as persistent nasal symptoms and/or an inadequate clinical response to nasal corticosteroids
- Coexisting persistent asthma and/or recurrent sinusitis/otitis
- A high pretest probability of allergic rhinitis and negative in vitro test results to suspected culprit allergens (because the sensitivity of skin testing is usually superior to that of in vitro testing)
- A patient's expressed desire to try to avoid the allergen rather than take medications to control symptoms

Blood tests (IgE immunoassays) — IgE immunoassays provide similar information as that obtained with allergen skin tests, although they are more expensive and less sensitive for the diagnosis of allergy to inhalant allergens compared with skin tests. However, in vitro testing can be useful when skin testing is not available or cannot be performed because

patients have extensive skin disease, cannot discontinue antihistamines or other interfering medications, are dermatographic, or have other issues that complicate skin testing [52,108,114,116-118]. The advantages of in vitro allergy testing as well as the sensitivity and specificity of IgE immunoassays are reviewed elsewhere. (See "Overview of in vitro allergy tests".)

Commercially available test panels intended for use by generalists typically detect IgE antibodies to common seasonal and perennial inhalant allergens [113]. However, this approach can be costly if excessive numbers of immunoassays are included or if the allergens selected are not relevant to the geographic area in question. Furthermore, such panel testing has the potential for false positives, and it can be difficult to interpret results. A logical approach would involve consulting an allergy expert in the area initially to identify a small number of important allergens for the area (eg, a few prominent pollens, animal danders, molds for dry environments, and dust mites for humid environments) [119].

Suggestive history with negative testing — Occasionally, a patient presents with a history and physical findings suggestive of allergic rhinitis, but skin testing and in vitro testing are negative. Most commonly, such patients have chronic nonallergic rhinitis, which has subtly different historical features and physical findings. (See 'Differential diagnosis' below.)

Another possibility is that the patient is producing allergen-specific IgE locally in the nasal tissues, but it is not reflected in the systemic circulation or skin. However, nasal allergen provocation tests are positive. This condition is sometimes referred to as "local allergic rhinitis," which may have either intermittent or persistent symptoms, and it is an area of increasing research interest [120]. Local allergic rhinitis is discussed separately. (See "Chronic nonallergic rhinitis".)

Fortunately, patients with either of these disorders may be managed similarly to those with allergic rhinitis. (See "Pharmacotherapy of allergic rhinitis", section on 'Summary and recommendations'.)

Uncommonly used tests — Nasal cytology and direct inhalational challenge with allergen are techniques that are largely limited to research settings.

- **Nasal cytology** – Nasal cytology is performed by some investigators to help differentiate rhinitis due to allergy from that due to infection, although it is relatively nonspecific and insensitive. Nasal secretions may be obtained with a cotton swab or by asking the patient to blow the nose onto waxed paper or cellophane. Wright stain of nasal secretions usually but not always reveals a predominance of eosinophils in cases of allergic rhinitis. By comparison, the presence of neutrophils suggests an infectious process.

  Nasal eosinophilia may also be seen in other conditions including:

  - Asthma without symptoms of nasal allergy
  - Nasal polyposis, with or without asthma and aspirin sensitivity (see "Aspirin-exacerbated respiratory disease")
  - Nonallergic rhinitis with eosinophilia syndrome (NARES), a syndrome of marked nasal eosinophilia and a propensity for nasal polyps but with negative allergic histories, negative skin tests, and no aspirin sensitivity (see 'Differential diagnosis' below)

- **Nasal challenge** – Although nasal allergen challenge can definitively establish the diagnosis, it is clinically impractical and rarely performed outside of research settings. Nasal challenge procedures are discussed elsewhere. (See "Chronic nonallergic rhinitis" and "Occupational rhinitis".)

Unproven diagnostic tests — There are several unproven or inappropriate diagnostic testing assays that are being offered by various types of providers with increasing frequency. These include cytotoxic testing, provocation neutralization testing, and specific or nonspecific immunoglobulin G (IgG) determinations [121]. The results of these methods are **not** useful for diagnosis or management.

DIFFERENTIAL DIAGNOSIS

Various other disorders can mimic allergic rhinitis or coexist with it in older children and adults.

Children under two years of age — Because allergic sensitization takes a few years to develop, other disorders should be considered in very young children who have persistent rhinitis symptoms. (See 'Natural history' above.)

These include adenoidal hypertrophy, acute or chronic sinusitis, congenital abnormalities (choanal atresia), foreign bodies, and nasal polyps. (See "The pediatric physical examination: HEENT", section on 'Nose'.)

Older children and adults — Rhinitis can result from either inflammatory or noninflammatory causes [122].

- **Acute infectious rhinitis** – Symptoms of the common cold vary from patient to patient, with rhinitis and nasal congestion being the most common. Nasal obstruction, rhinorrhea, and sneezing are usually present early in the course of the cold, although a sore or "scratchy" throat is frequently the most bothersome symptom on the first day of illness. The sore throat is usually short lived, and nasal symptoms predominate by the second and third day. Cough typically becomes troublesome on the fourth or fifth day of illness, by

which time the nasal symptoms are less severe. (See "The common cold in adults: Diagnosis and clinical features".)

Acute bacterial sinusitis develops in 0.5 to 2.5 percent of adult patients after viral upper respiratory tract infection. Viral sinusitis occurs much more frequently. The clinical presentation of the patient is of limited utility in distinguishing cases of pure viral rhinosinusitis from those with secondary bacterial infection. There appear to be **no** signs and symptoms of acute respiratory illness that are both sensitive and specific in making this distinction. (See "Acute sinusitis and rhinosinusitis in adults: Clinical manifestations and diagnosis".)

The remaining disorders discussed below result from chronic processes.

- **Chronic nonallergic rhinitis** – Approximately 50 percent of patients with chronic rhinitis have a component of nonallergic rhinitis [123]. Chronic nonallergic rhinitis is characterized by perennial symptoms and mild or absent nasal itching and sneezing. Patients with this disorder complain of chronic nasal congestion and/or rhinorrhea that is intensified by rapid changes in temperature and relative humidity, odors, or alcohol. They have little nasal itching or sneezing. However, headaches, anosmia, and sinusitis are common. A family history of allergy or allergic symptom triggers is uncommon. Negative skin tests to inhalant allergens are essentially diagnostic for a nonallergic rhinitis syndrome. Syndromes of nonallergic rhinitis include vasomotor rhinitis, gustatory rhinitis, and nonallergic rhinitis with nasal eosinophilia syndrome. These are reviewed in more detail separately. (See "Chronic nonallergic rhinitis".)

- **Chronic rhinosinusitis** – Chronic rhinosinusitis (CRS) is defined as an inflammatory condition involving the paranasal sinuses and linings of the nasal passages that lasts 12 weeks or longer despite attempts at medical management. CRS can coexist with allergic rhinitis. The diagnosis of CRS requires any **two** of the following symptoms, present for at least 12 weeks:

  - Anterior and/or posterior mucopurulent drainage
  - Nasal obstruction
  - Facial pain, pressure, and/or fullness
  - Decreased sense of smell

  Concomitant CRS should be considered when a patient with allergic rhinitis also has the above mentioned symptoms and fails to improve with treatment for allergic rhinitis. The diagnosis of CRS is discussed in greater detail elsewhere. (See "Chronic rhinosinusitis: Clinical manifestations, pathophysiology, and diagnosis".)

- **Rhinitis medicamentosa** – Rhinitis medicamentosa is a complication of vasoconstrictor nasal sprays (which may develop with as little as five days of use) or intranasal cocaine use.

Chronic nasal obstruction and nasal inflammation develop and are manifested as beefy red nasal membranes on physical examination. Diagnosis depends almost entirely on the appropriate history, characteristic physical examination findings, and positive response to treatment with topical nasal corticosteroids, which is usually required to withdraw successfully from the culprit medication [124-126]. (See "Chronic nonallergic rhinitis".)

- **Rhinitis due to systemic medications** – A variety of systemic medications can induce nasal symptoms. This is in contrast to rhinitis medicamentosa, which is a specific syndrome that results from the intranasal application of certain drugs.

  Classes of medications that can cause nasal symptoms include birth control pills, antihypertensive drugs (alpha-adrenergic blockers, beta-adrenergic blockers, angiotensin-converting enzyme [ACE] inhibitors), erectile dysfunction drugs, and nonsteroidal antiinflammatory drugs (NSAIDs). Psychiatric medications that have been implicated include chlorpromazine, thioridazine, perphenazine, chlordiazepoxide, amitriptyline, and alprazolam. Lastly, the immunosuppressants cyclosporine and mycophenolic acid can cause nasal symptoms [127]. Rhinitis symptoms caused by these medications generally subside within a few weeks of discontinuation.

- **Atrophic rhinitis** – Atrophic rhinitis is a syndrome of progressive atrophy of the nasal mucosa usually seen in older adults. This can develop in patients who have had multiple sinus surgeries, resulting in loss of normal mucociliary function. Such individuals report chronic nasal congestion that is refractory to common treatments for allergic rhinitis. One variant occurs in patients who have had multiple sinus surgeries, resulting in loss of normal mucociliary function. In resource-limited areas, another variant is associated with mucosal colonization with *Klebsiella ozaenae*, and these patients may perceive a persistent bad odor. This is discussed separately. (See "Atrophic rhinosinusitis".)

- **Rhinitis associated with hormonal changes** – Rhinitis of pregnancy and rhinitis of hypothyroidism reflect nasal obstruction that occurs on a hormonal basis. In these settings, the diagnosis is clinical and is supported by negative skin tests or improved symptoms upon resolution or treatment of the causative condition. (See "Clinical manifestations of hypothyroidism" and "Maternal adaptations to pregnancy: Dyspnea and other physiologic respiratory changes", section on 'Physiologic pulmonary changes in pregnancy'.)

- **Unilateral rhinitis or nasal polyps** – Unilateral rhinitis or nasal polyps are uncommon in uncomplicated allergic rhinitis. Unilateral rhinitis suggests the possibility of nasal obstruction by a foreign body, tumor, or polyp, and the presence of nasal polyps suggests nonallergic rhinitis with eosinophilia syndrome (NARES), chronic bacterial sinusitis, allergic fungal sinusitis, aspirin hypersensitivity, cystic fibrosis, or primary ciliary dyskinesia (immotile cilia syndrome). Fiberoptic rhinoscopy may be helpful in this setting. Increasing evidence also suggests that the histology of those with rhinitis with or without nasal polyps

are different, with eosinophils or neutrophils predominating in those with or without polyps, respectively [122]. However, allergic rhinosinusitis and NARES are both conditions characterized by nasal eosinophilia, but most patients with these conditions lack nasal polyps. (See "Chronic rhinosinusitis: Clinical manifestations, pathophysiology, and diagnosis" and "Fungal rhinosinusitis" and "Epidemiology and clinical manifestations of invasive aspergillosis" and "Cystic fibrosis: Clinical manifestations and diagnosis".)

- **Rhinitis with immunologic disorders** – A number of systemic autoimmune disorders present with nasal symptoms or can affect nasal mucosa. These include granulomatosis with polyangiitis and relapsing polychondritis:

  - The most common presenting symptoms of granulomatosis with polyangiitis include persistent rhinorrhea, purulent/bloody nasal discharge, oral and/or nasal ulcers, polyarthralgias, myalgias, or sinus pain.

  - With relapsing polychondritis, symptoms of stuffiness, crusting, rhinorrhea, and, on occasion, epistaxis may accompany nasal cartilage inflammation, which can also compromise olfaction. Cartilage destruction associated with sustained or recurrent episodes of inflammation can result in a characteristic saddle-nose deformity.

  Such disorders may therefore present with nasal symptoms, without evidence of systemic disease. These disorders are diagnosed based upon the combination of characteristic histologic and clinical findings. (See "Granulomatosis with polyangiitis and microscopic polyangiitis: Clinical manifestations and diagnosis" and "Clinical manifestations of relapsing polychondritis".)

LOCAL ALLERGIC RHINITIS

Local allergic rhinitis (LAR), formerly considered a phenotype of idiopathic nonallergic rhinitis or a subtype of nonallergic rhinitis with eosinophilia (NARES), is a subclassification of allergic rhinitis in which patients have a clinical presentation suggestive of allergic rhinitis but lack specific IgE (sIgE) detectable by allergen-specific skin prick or serum in vitro testing. However, sIgE may be detected in nasal lavage specimens. Allergen-specific nasal provocation testing is considered the gold standard to diagnose LAR in patients without systemic sIgE, although this is rarely done outside of research settings [128-130].

The immunopathogenesis of local rather than systemic production of sIgE in LAR is not well understood. Some investigators have speculated that sIgE may be present in sufficient quantity in affected patients to sensitize their nasal effector cells but insufficient quantity to sensitize cutaneous mast cells or be detected in serum [129,130].

SOCIETY GUIDELINE LINKS

Links to society and government-sponsored guidelines from selected countries and regions around the world are provided separately. (See "Society guideline links: Rhinitis".)

INFORMATION FOR PATIENTS

UpToDate offers two types of patient education materials, "The Basics" and "Beyond the Basics." The Basics patient education pieces are written in plain language, at the 5th to 6th grade reading level, and they answer the four or five key questions a patient might have about a given condition. These articles are best for patients who want a general overview and who prefer short, easy-to-read materials. Beyond the Basics patient education pieces are longer, more sophisticated, and more detailed. These articles are written at the 10th to 12th grade reading level and are best for patients who want in-depth information and are comfortable with some medical jargon.

Here are the patient education articles that are relevant to this topic. We encourage you to print or e-mail these topics to your patients. (You can also locate patient education articles on a variety of subjects by searching on "patient info" and the keyword(s) of interest.)

- Basics topics (see "Patient education: Environmental allergies in adults (The Basics)" and "Patient education: Environmental allergies in children (The Basics)" and "Patient education: Allergy skin testing (The Basics)")
- Beyond the Basics topic (see "Patient education: Allergic rhinitis (Beyond the Basics)")

SUMMARY

- **Clinical manifestations** – Allergic rhinitis is characterized by paroxysms of sneezing, rhinorrhea, nasal obstruction, postnasal drainage, and itching of the eyes, nose, and palate. More insidious effects of the disorder include fatigue, irritability, reduced performance at school and work, and depression. (See 'Introduction and terminology' above and 'Clinical manifestations' above.)
- **Epidemiology and economic impact** – Allergic rhinitis affects 10 to 30 percent of children and adults in the United States. The prevalence is increasing in industrialized countries worldwide, particularly in urban areas. Although frequently underdiagnosed and undertreated, allergic rhinitis imposes a significant economic burden as a result of reduced school performance and work productivity, clinician visits, expense of over-the-counter and prescription medications, and cost of treating related conditions, such as sinusitis and asthma. (See 'Epidemiology' above and 'Associated conditions' above.)

- **Subtypes** – Allergic rhinitis may be classified by temporal pattern (intermittent or persistent) and by severity (mild or moderate-severe) (table 1). (See 'Patterns of symptoms' above.)

- **Diagnosis** – The diagnosis of allergic rhinitis is made clinically based on a suggestive history (including presence of risk factors), characteristic symptoms and signs on physical examination, and (if indicated) the confirmed presence of allergen-specific immunoglobulin E (IgE). Symptoms should also be reproducible (by history) upon exposure to allergens to which the patient is sensitized. (See 'Diagnosis' above.)

- **Allergen-specific testing** – Identification of the specific allergens to which the patient is sensitive is not necessary for successful treatment in many cases. However, patients whose symptoms are severe or refractory to therapy should be referred to an allergy specialist for a more definitive evaluation. Properly performed skin testing is the best method for determining allergic sensitization. (See 'Allergen-specific testing' above.)

- **Differential diagnosis** – The differential diagnosis of allergic rhinitis includes acute and chronic rhinosinusitis (CRS), chronic nonallergic rhinitis, rhinitis medicamentosa, atrophic rhinitis, and rhinitis due to systemic medications. (See 'Differential diagnosis' above.)

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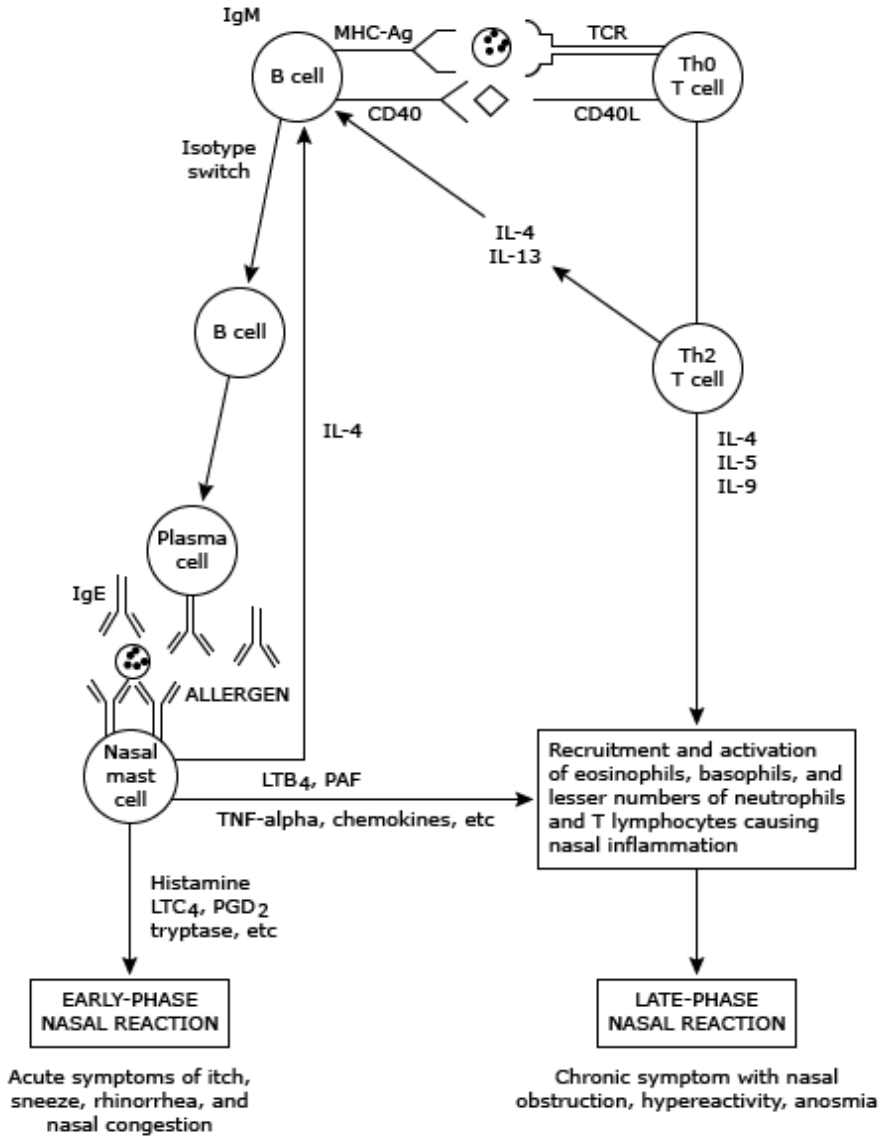
Topic 7525 Version 38.0

GRAPHICS

Figure 1: **Early and late phases of an allergic reaction in the nasal tissues**



This schematic shows pathways (including significant cells and cytokines) underlying early (left) and late (right) phase nasal reactions. This is a simplified overview of what is known about the immunopathogenesis of allergic rhinitis. However, the complexity of the process is appreciated in greater detail, providing information which may ultimately result in new therapies. First, proteases from inhaled allergens promote penetration of the nasal epithelial barrier and release of epithelial cytokines to include thymic stromal lymphopoietin, IL-25, and IL-33. These cytokines polarize dendritic cells to interact with lymphocytes promoting a Th2-type cascade of events as dendritic cells then process allergen and present it to naive T-cells and B-cells in lymph nodes. Allergen specific T-follicular helper cells and Th2 lymphocytes; and, type 2 innate lymphoid cells (with no T-cell receptor) respond by producing IL-4, IL-5, and IL-13 that facilitate B-lymphocyte isotype switches to IgE production and the resulting plasma cells produce allergen-specific IgE. That IgE binds to FceR1 on nasal mast cells. With repeat allergen exposure, nasal mucosal mast cells are activated and release mediators of immediate hypersensitivity. Nasal symptoms begin with mast cell mediator release of vasoactive amines and lipid mediators that produce the immediate allergic response. Cytokines are also released that subsequently produce the late phase allergic response. Those cytokines promote in-migration of

activated neutrophils, eosinophils, basophils and helper T- cells with increased IL-4, IL-5, IL-9 and IL-13 release that produces the nasal inflammation of the late phase response.

IgM: immunoglobulin M; IgE: immunoglobulin E; LTB_4: leukotriene B_4; PAF: platelet-activating factor; TNF-alpha: tumor necrosis factor-alpha; LTC_4: leukotriene C_4; PGD_2: prostaglandin D_2; MHC-Ag: major histocompatibility complex bound to antigen; TCR: T cell receptor; CD40L: CD40 ligand; Th0: undifferentiated T cell; IL: interleukin; Th2: T helper type 2.

Adapted with permission from: deShazo RD. In: Cecil Textbook of Medicine, 5th ed, Goldman (Ed), 2002. Copyright © 2002 Elsevier Science.

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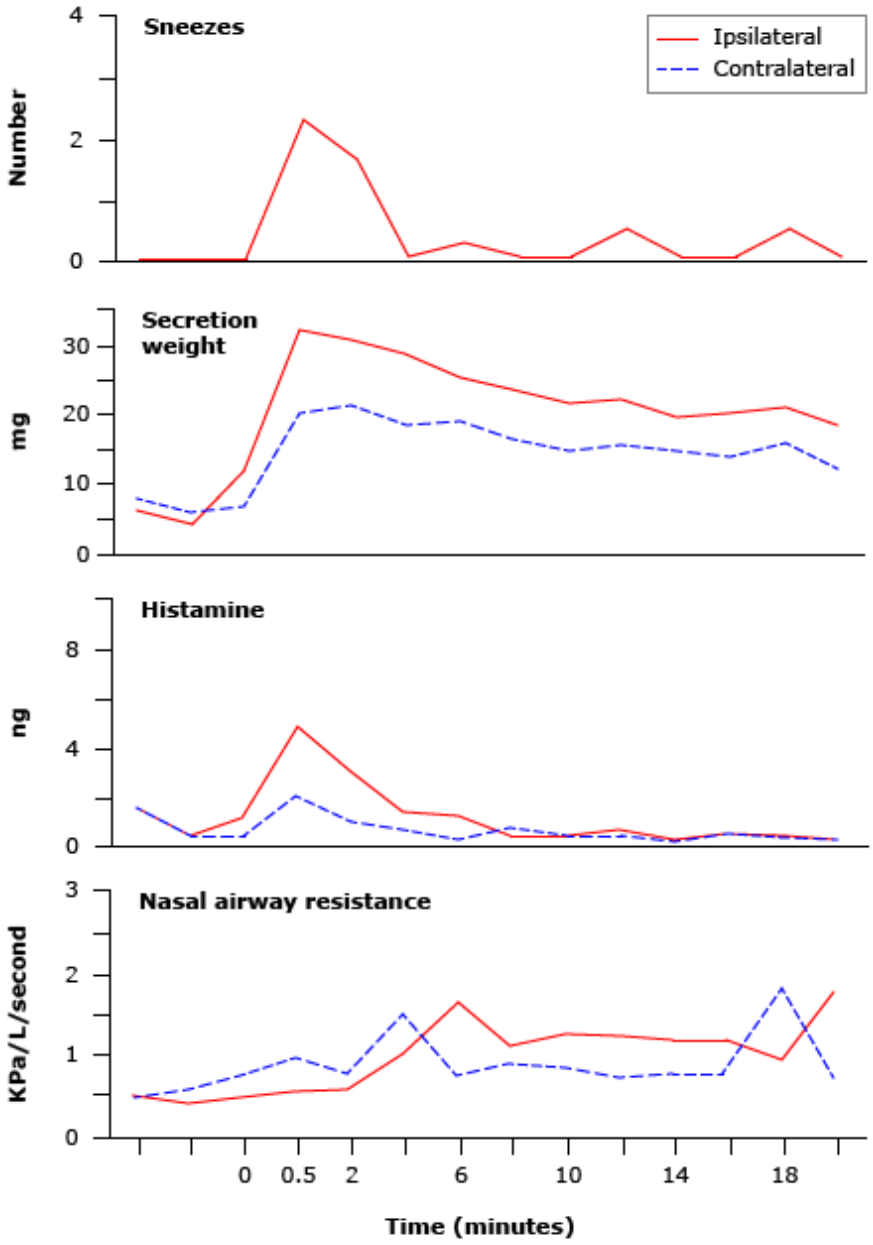
Figure 2: **Responses to nasal allergen challenge**



Responses to allergen challenge: Allergen is placed in the ipsilateral nostril (red solid lines). The graphs show the time course of histamine release and physiologic changes during the early-phase response. The time course of histamine concentration, symptoms (sneezing), and increases in nasal airway resistance are closely correlated.

Reproduced with permission from: Baroody FM, Naclerio RM. Chapter 48. In: Second Edition Clinical Immunology Principles and Practice, Volume 1, Rich RR, Fleisher TA, Shearer WT, et al (Eds), Mosby 2001. p.48.3. Copyright © 2001 Elsevier.

Graphic 65419 Version 5.0

Table 1: **Classification of allergic rhinitis**

"Intermittent" means that the symptoms are present: ▪ Less than 4 days a week ▪ Or for less than 4 weeks
"Persistent" means that the symptoms are present: ▪ More than 4 days a week ▪ And for more than 4 weeks
"Mild" means that none of the following items are present: ▪ Sleep disturbance ▪ Impairment of daily activities, leisure, and/or sport ▪ Impairment of school or work ▪ Troublesome symptoms
"Moderate-severe" means that one or more of the following items are present: ▪ Sleep disturbance ▪ Impairment of daily activities, leisure, and/or sport ▪ Impairment of school or work ▪ Troublesome symptoms

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Graphic 85849 Version 3.0

Figure 3: **Peak pollen periods in the United States**

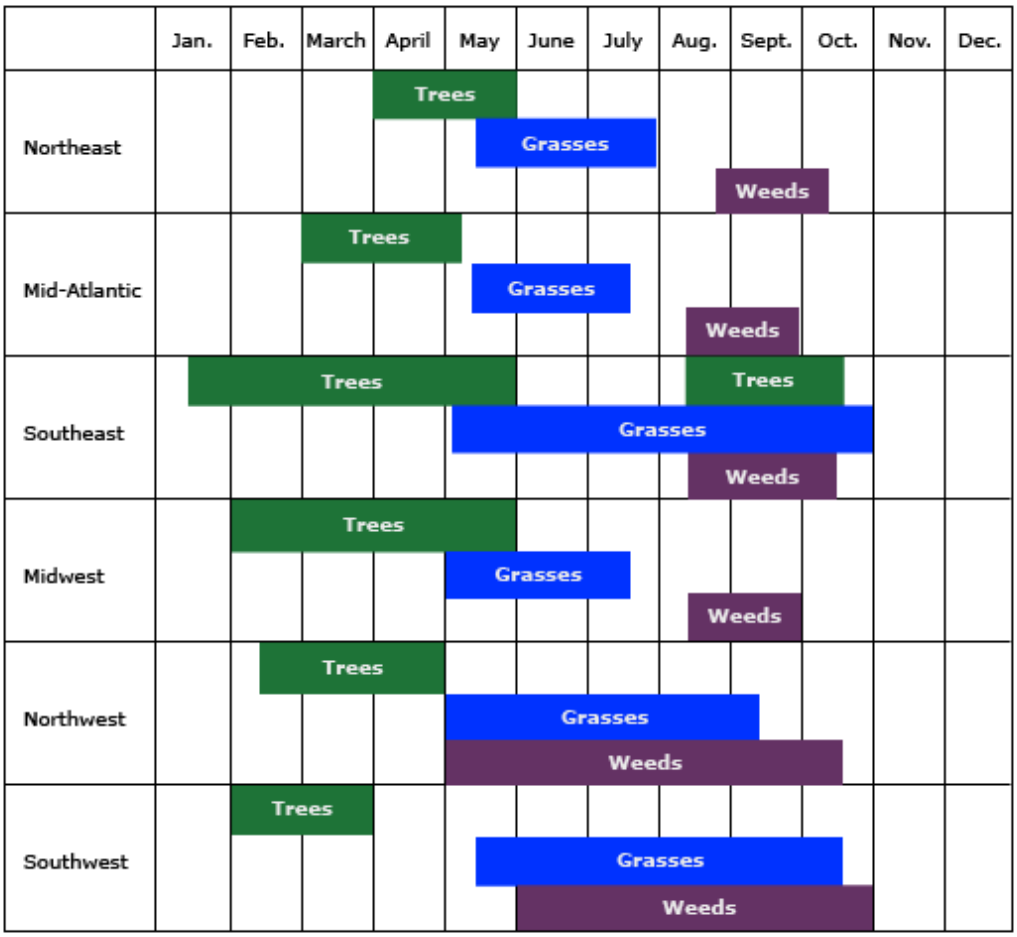

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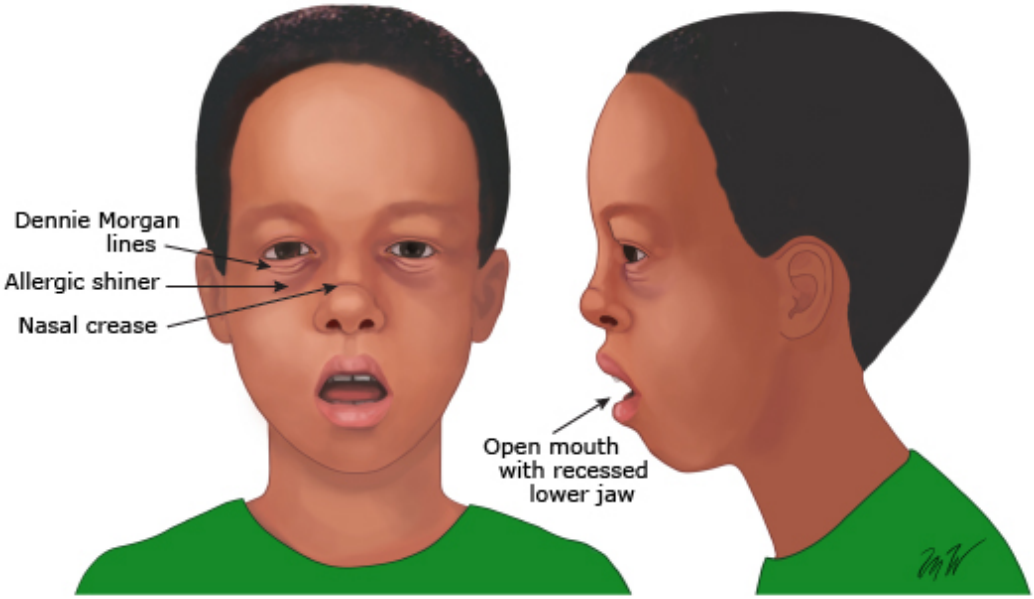
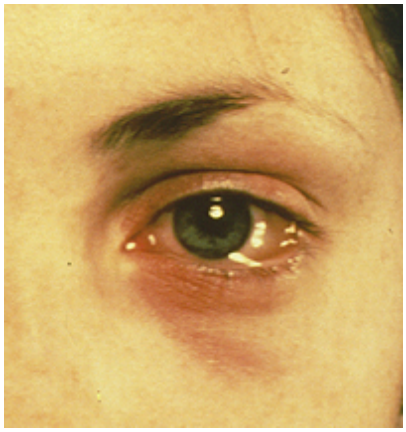
Figure 4: **Child with allergies**



Children with allergic rhinitis may display "allergic shiners" and "adenoid facies," resulting from longstanding mouth-breathing. The features of adenoid facies include a long, narrow face; narrow, upturned nose; short upper lip; exposed upper incisors; recessed lower jaw; and forward head posture.

Graphic 66393 Version 6.0

Picture 1: **Acute seasonal conjunctivitis**



Acute seasonal conjunctivitis classically presents with eyelid and conjunctival erythema and swelling. It is almost always bilateral.

Graphic 57240 Version 2.0

Table 2: **Sample questions* for the diagnosis and initial assessment of asthma**

A "yes" answer to any question suggests that an asthma diagnosis is likely.
In the past 12 months, have you¶...
Had a sudden severe episode or recurrent episodes of coughing, wheezing (high-pitched whistling sounds when breathing out), chest tightness, or shortness of breath?
Had colds that "go to the chest" or take more than 10 days to get over?
Had coughing, wheezing, or shortness of breath during a particular season or time of the year?
Had coughing, wheezing, or shortness of breath in certain places or when exposed to certain things (eg, animals, tobacco smoke, perfumes)?
Used any medications that help you breathe better? How often?
Had symptoms relieved when the medications are used?
In the past 4 weeks, have you¶ had coughing, wheezing, or shortness of breath...
At night that has awakened you?
Upon awakening?
After running, moderate exercise, or other physical activity?

* These questions are examples and do not represent a standardized assessment or diagnostic instrument. The validity and reliability of these questions have not been assessed.

¶ Or "your child," if a parent/caregiver is answering the questions for a child.

Reproduced from: National Asthma Education and Prevention Program, Third Expert Panel on the Diagnosis and Management of Asthma. Expert panel report 3: Guidelines for the diagnosis and management of asthma. Publication no. 08-4051, National Heart, Lung, and Blood Institute, Bathesda, MD 2007.

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